Bank Management System

Code and Branch Coverage Report
Generated: 2026-11-11 18:53:45

Code Coverage

Line coverage is measured in C++ implementation sources; the interactive src/main.cpp entry point is excluded.

Source file	Covered	Total	Coverage
src/infra/AccountRepository.cpp	85	102	83.33%
src/infra/AuditLogger.cpp	20	20	100.00%
src/infra/Config.cpp	37	43	86.05%
src/infra/CredentialStore.cpp	56	63	88.89%
src/infra/CustomerAccountStore.cpp	215	218	98.62%
src/infra/DataIntegrityValidator.cpp	156	158	98.73%
src/infra/FileUtil.cpp	34	38	89.47%
src/infra/InputValidator.cpp	58	59	98.31%
src/infra/Journal.cpp	185	187	98.93%
src/infra/LoginService.cpp	27	27	100.00%
src/infra/PasswordHasher.cpp	78	81	96.30%
src/infra/TimeUtil.cpp	9	9	100.00%
src/infra/TransactionRepository.cpp	101	116	87.07%
src/scaffold_placeholder.cpp	0	1	0.00%
src/service/AccountClosureService.cpp	27	27	100.00%
src/service/AccountDetailsService.cpp	14	15	93.33%
src/service/AccountNumberGenerator.cpp	7	7	100.00%
src/service/AccountService.cpp	94	97	96.91%
src/service/AccountUnlockService.cpp	17	21	80.95%
src/service/AuditLogReviewService.cpp	42	45	93.33%
src/service/AuthService.cpp	17	19	89.47%
src/service/CustomerSearchService.cpp	56	60	93.33%
src/service/CustomerUpdateService.cpp	21	25	84.00%
src/service/RbacGuard.cpp	6	6	100.00%
src/service/SessionManager.cpp	25	25	100.00%

Source file	Covered	Total	Coverage
src/service/SystemStatusService.cpp	23	23	100.00%
src/service/TransactionService.cpp	341	362	94.20%
TOTAL	**1751**	**1854**	**94.44%**

Branch Coverage

Branch coverage counts GCC-reported non-exception branch outcomes in C++ implementation sources; headers, tests, src/main.cpp, and exception-only edges are excluded. An outcome is covered when taken at least once.

Source file	Covered	Total	Coverage
src/infra/AccountRepository.cpp	90	124	72.58%
src/infra/AuditLogger.cpp	35	37	94.59%
src/infra/Config.cpp	30	42	71.43%
src/infra/CredentialStore.cpp	63	75	84.00%
src/infra/CustomerAccountStore.cpp	213	222	95.95%
src/infra/DataIntegrityValidator.cpp	201	212	94.81%
src/infra/FileUtil.cpp	35	38	92.11%
src/infra/InputValidator.cpp	61	65	93.85%
src/infra/Journal.cpp	202	220	91.82%
src/infra/LoginService.cpp	42	58	72.41%
src/infra/PasswordHasher.cpp	43	54	79.63%
src/infra/TimeUtil.cpp	4	4	100.00%
src/infra/TransactionRepository.cpp	130	156	83.33%
src/scaffold_placeholder.cpp	0	0	0.00%
src/service/AccountClosureService.cpp	27	28	96.43%
src/service/AccountDetailsService.cpp	15	67	22.39%
src/service/AccountNumberGenerator.cpp	10	10	100.00%
src/service/AccountService.cpp	107	122	87.70%
src/service/AccountUnlockService.cpp	12	15	80.00%
src/service/AuditLogReviewService.cpp	38	51	74.51%
src/service/AuthService.cpp	15	22	68.18%
src/service/CustomerSearchService.cpp	34	38	89.47%
src/service/CustomerUpdateService.cpp	23	34	67.65%
src/service/RbacGuard.cpp	4	4	100.00%
src/service/SessionManager.cpp	6	6	100.00%

Source file	Covered	Total	Coverage
src/service/SystemStatusService.cpp	11	11	100.00%
src/service/TransactionService.cpp	415	490	84.69%
TOTAL	**1866**	**2205**	**84.63%**

Coverage was collected with GCC gcov after the instrumented CTest run. A missing or unmeasured source file is reported as an error instead of being silently omitted.